Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2019-2020
Semester: Fall
Exam Date: 29/12/2019

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (MTHN103) Differential Equations and Transforms **Spec:** Comp. and Comm.
Examiners: Assoc. Prof. Dr. Ashraf Taha and Prof. Dr. Aly Essawi **Time:** 2 hours

Number of Pages: 2 Number of Questions: 4 Attempt all questions

Question 1 (10 points)

Solve the following differential equations:

a) $y\frac{dy}{dx}=\frac{1}{x}\left(\ln\frac{x}{y}+\ln y\right).$ (2 points)

b) $x\frac{dy}{dx}-y=xy^{7}.$ (2 points)

c) $\left(3x^{2}+y\cos(x)\right)dx+\left(\sin(x)-4y^{3}\right)dy=0.$ (3 points)

d) $\frac{dy}{dx}=\frac{x-y+1}{x-y-1}.$ (3 points)

Question 2 (10 points)

Find the general solution of the following differential equations:

a) $\frac{d^{3}y}{dx^{3}}-3\frac{d^{2}y}{dx^{2}}+6\frac{dy}{dx}-4y=0.$ (3 points)

b) $\frac{d^{2}y}{dx^{2}}+y=\sec(x)\csc(x).$ (3 points)

c) $\frac{d^{2}y}{dx^{2}}-\frac{dy}{dx}=5x+e^{3x}.$ (4 points)

Question 3 (12 points)

a) ***Solve the following differential equation using Laplace transform:***

$\frac{d^{2}y}{dt^{2}}-3\frac{dy}{dt}+2y=0;\qquad y(0)=0,y'(0)=1.$ (3 points)

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b) ***Find the Laplace transform of the following functions:***

i. $f(t) = u_2(t)\, e^3 \sin(t-2) + \sin(t)\cos(2t).$ (3 points)

ii. $f(t) = t^7\ e^{2t} + \int_0^t \sinh\tau\ \ d\tau.$ (2 points)

c) ***Find the inverse Laplace transform of the following functions:***

i. $F(s) = e^{-3s}\left(\frac{s}{s^2+10s+5}\right).$ (2 points)

ii. $F(s) = \ln\left(\frac{s+1}{s}\right).$ (2 points)

Question 4 (8 points)

a) ***Represent the following function*** $f(x) = x^4,\ \ 0 \le x \le \pi$ ***as Fourier Sine Series*** (3 points)

b) ***Prove the recurrence relations*** $xP_n(x) = \frac{n+1}{2n+1}P_{n+1}(x) + \frac{n}{2n+1}P_{n-1}(x).$ (3 points)

c) ***Use the above Prove to Evaluate the integral*** $\int_{-1}^{1} x\, P_5(x) P_4(x)\ \ dx.$ (2 points)

GOOD LUCK